Computational Mathematics

Sheet II.1 — HT25

1. Prove by induction that if Euclid's algorithm requires N divisions to compute the greatest common divisor of the pair $a > b, a, b \in \mathbb{N}$, then $a \geq F_{N+2}$ and $b \geq F_{N+1}$, where F_i denotes the ith Fibonacci number ($F_1 = 1, F_2 = 1, F_{i+2} = F_{i+1} + F_i$).

2. (a) Given $a, b, c \in \mathbb{Z}$, implement the extended Euclidean algorithm to find a solution (x, y) (if any exists) of

$$ax + by = c.$$

Use your code to find a solution to $993319x + 999331y = 939193$.

(b) The Rivest–Shamir–Adleman (RSA) cryptographic system is one of the most important protocols for securely exchanging messages. In most cryptographic systems, the key used for encryption and decryption are the same; in RSA, the key used for encryption is public, but the key used for decryption is kept secret.

The public key consists of two natural numbers, n and e. Here $n = pq$ where p and q are primes; p and q are kept secret. Knowledge of p and q lets us compute the *Carmichael totient function*

$$\lambda(n) = \frac{(p-1)(q-1)}{\gcd(p-1, q-1)}.$$

The public key must satisfy $1 < e < \lambda(n)$ and $\gcd(e, \lambda(n)) = 1$.

The associated private key d satisfies

$$de \equiv 1 \mod \lambda(n).$$

Using your implementation in part (a), compute the private key d for the public key $n = 137 \cdot 173, e = 7$.

(c) *(Optional)* encrypt the message $m = 21051$ using the formula

$$c \equiv m^e \mod n$$

and decrypt the resulting ciphertext c using

$$m \equiv c^d \mod n.$$

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3. Classify the order of convergence of the following sequences:
   (a) $a_n = 2^{-n}$.
   (b) $a_n = (n+1)^{-1}$.
   (c) $a_n = 2^{-2^n}$.

   For each sequence, plot a_n from $n = 1$ up to N, where $N = 100$ for (a) and (b) and $N = 10$ for (c), using a logarithmic scale on the y-axis.

4. Does Banach's contraction mapping theorem apply to the following continuous real-valued functions g with domain of definition X? Justify your answers.
   (a) $X = \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right], \quad g(x) = \sin x + x$.
   (b) $X = [0, 2], \quad g(x) = x^3 + 3x^2 - 3$.
   (c) $X = [1, \infty), \quad g(x) = x + x^{-1}$.

5. (a) Let $a_n \to a$ be a convergent sequence of real numbers. Let $e_n := |a_n - a|$ be its associated sequence of errors. Assume that the error can be described as

$$e_{n+1} \approx M e_n^q$$

   for constants M and q.

   Given three successive terms e_n, e_{n+1}, e_{n+2} in the sequence of errors, describe a procedure to estimate M and q.

   (b) Consider the Leibniz formula for π:

$$\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1},$$

   and let a_n be its sequence of partial sums.

   Estimate M and q for this sequence, starting from e_n with $n = 10,000$, and thereby state a conjecture about its convergence order.

   (c) Recall that the Aitken acceleration $\tilde{a}_n$ of a sequence a_n is given by

$$\tilde{a}_n = a_n - \frac{(a_{n+1} - a_n)^2}{a_{n+2} - 2a_{n+1} + a_n}.$$

   Plot the sequence of errors up to $n = 10,000$ of the partial sums a_n of the Leibniz formula for π, as well as its Aitken acceleration $\tilde{a}_n$, and the iterated Aitken acceleration $\tilde{\tilde{a}}_n$.

   (d) What is the estimated order of convergence of $\tilde{a}_n$ and $\tilde{\tilde{a}}_n$? Comment on your observations.

   [*Hint: you may wish to use higher-precision arithmetic for this question. Use* `mpmath` *with* `mpmath.mp.dps = 100`.]

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a $$\lim_{n \to \infty} \frac{2^{-(n+1)}}{(2^{-n})^q}$$

let $q = 1$ $\Rightarrow \lim_{n \to \infty} = \frac{1}{2}$ Order of convergence $= 1$.

$$\lim_{n \to \infty} \frac{(n+1)^{-1}}{n^{-q}} = \lim_{n \to \infty} \frac{n^q}{n+1} = 1 \text{ if } q = 1$$

$$\lim_{n \to \infty} \frac{2^{-(2^{n+1})}}{(2^{-2^n})^q} = \frac{2^{-2 \times 2^n}}{2^{-q \times 2^n}} = 1 \text{ if } q = 2.$$

4. Does Banach's contraction mapping theorem apply to the following continuous real-valued functions g with domain of definition X? Justify your answers.

(a) $X = \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$, $\quad g(x) = \sin x + x$.

(b) $X = [0, 2]$, $\quad g(x) = x^3 + 3x^2 - 3$.

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a)

$|g'(x)| = |\cos x + 1| \leq |1|$ over X

$\Rightarrow$ a) is a contraction

b)

$g'(x) = 3x^2 + 6x$

$|g'(2)| > 1 \Rightarrow$ b is not a contraction.

c) $g'(x) = 1 - x^{-2}$

$\Rightarrow |g'(x)| \leq 1 \quad \forall x$ so this is a contraction.

So the CMT holds for a and c.

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(d) What is the estimated order of convergence of $\tilde{a}_n$ and $\tilde{\tilde{a}}_n$? Comment on your observations.

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$$e_{n+2} = M e_{n+1}^q$$

$$= M(M e_n^q)^q$$

$$e_{n+2} = M^{q+1} e_n^{q^2}$$

$$e_{n+1} = M e_n^q$$

$$\ln(e_{n+2}) = (q+1)\ln M + q^2 \ln(e_n)$$

$$\ln(e_{n+1}) = \ln M + q \ln e_n$$

$$\ln(e_{n+2}) = (q+1)\left(\ln(e_{n+1}) - q\ln(e_n)\right) + q^2\ln(e_n)$$

$$\Rightarrow \ln(e_{n+2}) = (q+1)\ln(e_{n+1}) - q\ln(e_n)$$

$$\frac{\ln(e_{n+2}) - \ln(e_{n+1})}{\ln(e_{n+1}) - \ln(e_n)} = q$$

$$M = \frac{e_{n+1}}{e_n^q}$$